

Notes on lecture 2

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1 The vector product

Consider two vectors $\vec{a}$ and $\vec{b}$. Recall that $\vec{a}$ and $\vec{b}$ are orthogonal if and only if $\vec{a} \cdot \vec{b} = 0$. Suppose

$$\vec{a} = \langle a_1, a_2, a_3 \rangle$$

and

$$\vec{b} = \langle b_1, b_2, b_3 \rangle.$$

Suppose we are seeking a vector $\vec{c} = \langle c_1, c_2, c_3 \rangle$ such that $\vec{c}$ is orthogonal to both $\vec{a}$ and $\vec{b}$. Then we must have:

$$\vec{a} \cdot \vec{c} = 0 \text{ and } \vec{b} \cdot \vec{c} = 0.$$

That is, we must have:

$$a_1c_1 + a_2c_2 + a_3c_3 = 0$$

and

$$b_1c_1 + b_2c_2 + b_3c_3 = 0.$$

We want to find out c_1, c_2, c_3 . But since we have only two equations in three unknowns, we cannot solve the above system of equations uniquely. However, observe that the above system can be rewritten as

$$a_1c_1 + a_2c_2 = -a_3c_3$$

$$b_1c_1 + b_2c_2 = -b_3c_3.$$

Suppose we are trying to solve these equations only for c_1 and c_2 . Then we can simply use Cramer's rule (provided $\begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \neq 0$). So assuming that

$\begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \neq 0$, we have:

$$c_1 = \frac{\begin{vmatrix} -a_3c_3 & a_2 \\ -b_3c_3 & b_2 \end{vmatrix}}{\begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix}} \text{ and } c_2 = \frac{\begin{vmatrix} a_1 & -a_3c_3 \\ b_1 & -b_3c_3 \end{vmatrix}}{\begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix}}.$$

This implies that

$$c_1 = \frac{\begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} c_3}{\begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix}} \text{ and } c_2 = \frac{\begin{vmatrix} a_3 & a_1 \\ b_3 & b_1 \end{vmatrix} c_3}{\begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix}}.$$

Or in other words, we have:

$$\frac{c_1}{\begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix}} = \frac{c_3}{\begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix}} = \frac{c_2}{\begin{vmatrix} a_3 & a_1 \\ b_3 & b_1 \end{vmatrix}} = \lambda(\text{say}),$$

where λ is some nonzero scalar [Nonzero because if $\lambda = 0$, then we will get $\vec{c}$ to be the zero vector.]. Then

$$\vec{c} = \lambda[(a_2b_3 - a_3b_2)\hat{i} + (a_3b_1 - a_1b_3)\hat{j} + (a_1b_2 - a_2b_1)\hat{k}].$$

If we take $\lambda = 1$, then we will get

$$\vec{c} = (a_2b_3 - a_3b_2)\hat{i} + (a_3b_1 - a_1b_3)\hat{j} + (a_1b_2 - a_2b_1)\hat{k},$$

which is going to be orthogonal to both $\vec{a}$ and $\vec{b}$. The vector $\vec{c}$ (so obtained by taking $\lambda = 1$) is called the **vector product** or the **cross product** of $\vec{a}$ and $\vec{b}$. It is denoted by $\vec{a} \times \vec{b}$.

Remark 1.0.1. Observe that

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}.$$

This brings us to the concept of a right hand system. If we consider the right hand as $\vec{a}$ and the right palm as $\vec{b}$, then the vector $\vec{a} \times \vec{b}$ is a vector in the direction

of the thumb (You may refer to Figure 1.0.1.). If we invert the right hand, then we are going to get a vector (along the thumb) which is in a direction opposite to the direction of the vector $\vec{a} \times \vec{b}$. Thus we have, $\vec{a} \times \vec{b} = -(\vec{b} \times \vec{a})$ (although $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$).

Remark 1.0.2. If $\vec{a}$ and $\vec{b}$ are parallel, then $\vec{b} = \mu\vec{a}$ for some scalar μ . Then

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ \mu a_1 & \mu a_2 & \mu a_3 \end{vmatrix} = \vec{0}.$$

Remark 1.0.3. Observe that $\vec{a} \times \vec{b}$ is always a vector, not a scalar.

Theorem 1.0.4. $|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}| \sin \theta$, where θ is the angle between $\vec{a}$ and $\vec{b}$.

PROOF: Let $\vec{a} = \langle a_1, a_2, a_3 \rangle$ and $\vec{b} = \langle b_1, b_2, b_3 \rangle$. Then

$$\vec{a} \times \vec{b} = \langle a_2b_3 - a_3b_2, a_3b_1 - a_1b_3, a_1b_2 - a_2b_1 \rangle$$

and

$$\begin{aligned} |\vec{a} \times \vec{b}|^2 &= (a_2b_3 - a_3b_2)^2 + (a_3b_1 - a_1b_3)^2 + (a_1b_2 - a_2b_1)^2 \\ &= (a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2) - (a_1b_1 + a_2b_2 + a_3b_3)^2 \text{ (after simplification)} \\ &= |\vec{a}|^2|\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2 \\ &= |\vec{a}|^2|\vec{b}|^2 - |\vec{a}|^2|\vec{b}|^2 \cos^2 \theta \\ &= |\vec{a}|^2|\vec{b}|^2(1 - \cos^2 \theta) \\ &= |\vec{a}|^2|\vec{b}|^2 \sin^2 \theta. \end{aligned}$$

□

Remark 1.0.5. Let $\hat{n}$ denote a unit vector along $\vec{a} \times \vec{b}$, then

$$\vec{a} \times \vec{b} = |\vec{a} \times \vec{b}| \hat{n} = (|\vec{a}||\vec{b}| \sin \theta) \hat{n}.$$

Figure 1.0.1 explains this fact.

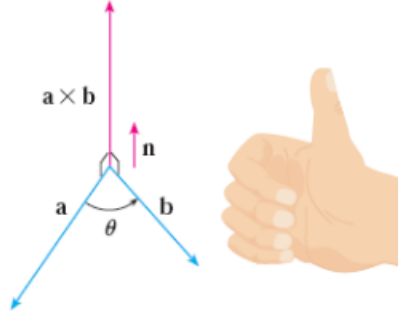


Figure 1.0.1: The vector product

1.1 Geometric interpretation of the vector product

Consider two vectors $\vec{a}$ and $\vec{b}$. Using parallel displacement (if needed), we can assume without loss of generality that both these vectors have the same initial point. Complete the parallelogram formed by these two vectors. Then the area of the parallelogram formed by the vectors $\vec{a}$ and $\vec{b}$ equals the length of the vector $\vec{a}$ times the length ($|\vec{b}| \sin \theta$, where θ denotes the angle between $\vec{a}$ and $\vec{b}$) of the perpendicular from the terminal point of $\vec{b}$ onto the vector $\vec{a}$. So we have:

$$\begin{aligned} &\text{The area of the parallelogram formed by } \vec{a} \text{ and } \vec{b} \\ &= |\vec{a}| |\vec{b}| \sin \theta = |\vec{a} \times \vec{b}|. \end{aligned}$$

Figure 1.1.1 provides an illustration.

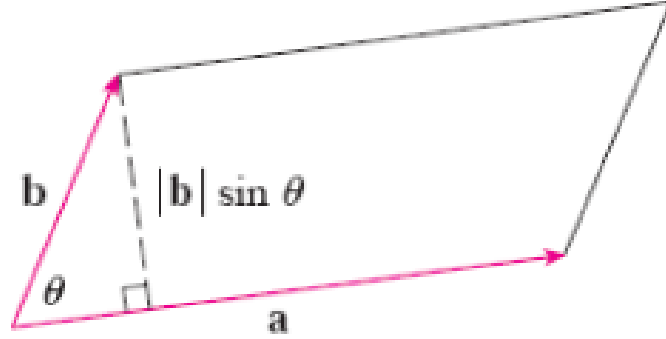


Figure 1.1.1: Geometric interpretation of the vector product

2 Scalar triple product

Consider 3 vectors $\vec{a} = \langle a_1, a_2, a_3 \rangle$, $\vec{b} = \langle b_1, b_2, b_3 \rangle$ and $\vec{c} = \langle c_1, c_2, c_3 \rangle$. Then

$$\vec{b} \times \vec{c} = \langle b_2c_3 - b_3c_2, b_3c_1 - b_1c_3, b_1c_2 - b_2c_1 \rangle$$

and

$$\vec{a} \cdot (\vec{b} \times \vec{c}) = a_1(b_2c_3 - b_3c_2) + a_2(b_3c_1 - b_1c_3) + a_3(b_1c_2 - b_2c_1)$$

$$= \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \rightarrow \text{This is a scalar.}$$

This scalar $\vec{a} \cdot (\vec{b} \times \vec{c})$ is called the **scalar triple product** of $\vec{a}$, $\vec{b}$ and $\vec{c}$.

2.1 Geometric interpretation of the scalar triple product

We know that the area of the parallelogram formed by the vectors $\vec{b}$ and $\vec{c}$ equals $|\vec{b} \times \vec{c}|$. Now, if we consider the parallelepiped whose edges are $\vec{a}$, $\vec{b}$, $\vec{c}$, then the volume of that parallelepiped is given by $h|\vec{b} \times \vec{c}|$, where h is the length of the perpendicular from the terminal point of $\vec{a}$ over the plane in which the vectors $\vec{b}$ and $\vec{c}$ lie. The cross product $\vec{b} \times \vec{c}$ is a vector perpendicular to both $\vec{b}$ and $\vec{c}$. Look at Figure 2.1.1 for an illustration. Suppose $\hat{n}$ is the

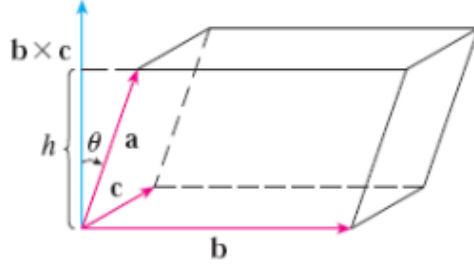


Figure 2.1.1: Geometric interpretation of the scalar triple product

unit vector along $\vec{b} \times \vec{c}$. If θ is the angle between $\hat{n}$ and $\vec{a}$, then $h = |\vec{a}| \cos \theta$. Then the volume of the parallelepiped equals

$$\begin{aligned} &= h|\vec{b} \times \vec{c}| \\ &= |\vec{a}||\vec{b} \times \vec{c}| \cos \theta \\ &= |\vec{a} \cdot (\vec{b} \times \vec{c})| \end{aligned}$$

3 Equation of a straight line in space

We want to find out the equation of a straight line L in space which passes through a fixed point. Suppose $P_0(x_0, y_0, z_0)$ is that fixed point. Also, it is given that the straight line which passes through the point P_0 is parallel to a given vector $\vec{v}$. Figure 3.0.1 illustrates the scenario.

We will take a variable point $P(x, y, z)$ on L and then we will join OP_0 and OP (here O denotes the origin). Let $\overrightarrow{OP_0} = \vec{r}_0$ and $\overrightarrow{OP} = \vec{r}$. Let us assume that $\vec{v} = \langle a, b, c \rangle$. From the triangle OP_0P , we can see that

$$\overrightarrow{OP} = \overrightarrow{OP_0} + \overrightarrow{P_0P},$$

or in other words,

$$\vec{r} = \vec{r}_0 + t\vec{v}, \quad (\text{since } \overrightarrow{P_0P} \text{ is parallel to } \vec{v})$$

where t is some scalar. The equation $\vec{r} = \vec{r}_0 + t\vec{v}$ is the **vector equation of the required straight line L** . In other words, the vector equation of

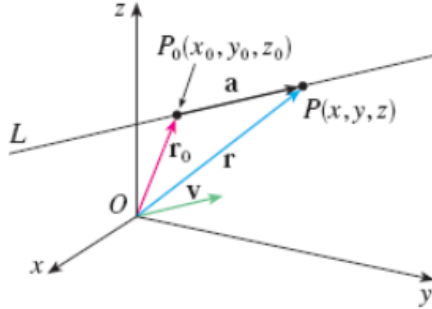


Figure 3.0.1: Equation of a straight line in space

the straight line L is given by $\langle x, y, z \rangle = \langle x_0, y_0, z_0 \rangle + t \langle a, b, c \rangle$. If $t > 0$, then the direction of $\overrightarrow{P_0P}$ is going to be along the direction of $\vec{v}$. In that case, the point P is going to lie on the right side of the point P_0 . If $t < 0$, then the direction of $\overrightarrow{P_0P}$ is going to be opposite to the direction of $\vec{v}$. In that case, the point P is going to lie on the left side of the point P_0 . If $t = 0$, then $P = P_0$.

The vector equation can be rewritten as the following set of equations:

$$x = x_0 + at, \quad y = y_0 + bt, \quad z = z_0 + ct. \quad (A)$$

The set of equations (A) is called the **parametric form** of the equation of the straight line L . It is called the parametric form because the values of x, y, z are dependent on the parameter t . Each value of the parameter t gives a point (x, y, z) on the line L .

Now, if we eliminate t from the set of equations (A), we get:

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}. \quad (B)$$

You can see that there is a symmetry in all the 3 ratios in equation (B) above. That's why equation (B) is called the **symmetric form** of the equation of the straight line L . Observe that in the symmetric form (B), the denominators of all those 3 ratios are a, b, c , which are nothing but the components of the vector $\vec{v}$ to which the given straight line L is parallel to.

Let us now go back to what we had learnt earlier about direction angles and direction cosines. Let α, β, γ be the angles made by the vector $\vec{v}$ with the x, y and z -axes respectively. These angles α, β, γ are called the direction angles of $\vec{v}$. Then $\cos \alpha, \cos \beta, \cos \gamma$ are the direction cosines of $\vec{v}$. We also know from earlier lectures that

$$\langle a, b, c \rangle = |\vec{v}| \langle \cos \alpha, \cos \beta, \cos \gamma \rangle . \quad (C)$$

Since $\vec{v}$ is parallel to the straight line L , therefore the direction angles and direction cosines of the straight line L are going to be the same as those of $\vec{v}$.

Now consider the portion P_0P of the straight line L (see Figure 3.0.1). If we make this portion of the line L into a vector (namely, the vector $\overrightarrow{P_0P}$), then obviously (since L is parallel to $\vec{v}$) the vector $\overrightarrow{P_0P}$ is going to be a scalar multiple of the vector $\vec{v} = \langle a, b, c \rangle$, say it is given by $t \langle a, b, c \rangle$. Then we have:

$$t \langle a, b, c \rangle = t|\vec{v}| \langle \cos \alpha, \cos \beta, \cos \gamma \rangle . \quad (D)$$

Observe that the numbers a, b, c are not exactly the direction cosines of L , but they are some scalar multiples of the direction cosines $\cos \alpha, \cos \beta, \cos \gamma$ of L . That is why the numbers a, b, c are called **direction ratios** of L .

Remark 3.0.1. The direction ratios of any two parallel lines are proportional.

Remark 3.0.2. Since any vector parallel to $\vec{v}$ could also be used for all the above discussion, we say that any 3 numbers proportional to a, b, c can also be used as a set of direction ratios of the straight line L .

Example 3.0.3. Find the equation of the straight line L passing through the point $P_0(5, 1, 3)$ and parallel to $\hat{i} + 4\hat{j} - 2\hat{k}$. Also find any two other points on that straight line. Please see Figure 3.0.2 for an illustration. Here we have, $\vec{v} = \langle 1, 4, -2 \rangle$ and the point P_0 is given by $(5, 1, 3)$. So the equation of the straight line L is going to be

$$\vec{r} = \langle 5, 1, 3 \rangle + t \langle 1, 4, -2 \rangle .$$

In symmetric form, the same equation is given by:

$$\frac{x-5}{1} = \frac{y-1}{4} = \frac{z-3}{-2} .$$

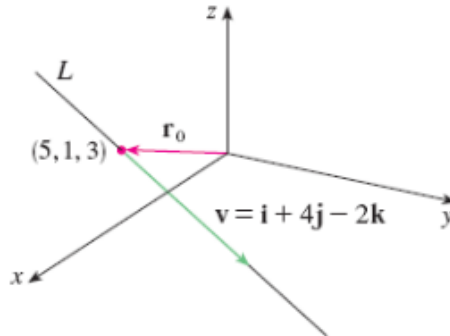


Figure 3.0.2: Figure for Example 3.0.3

Now, depending upon the value of the parameter t , we will be able to find all points lying on the straight line L . If $t > 0$, then the point corresponding to this value of t will lie on the right side of P_0 . Similarly, if $t < 0$, then the point corresponding to this value of t will lie on the left side of P_0 . Please see Figure 3.0.3 for an illustration.

Taking $t = 1$, we get $\vec{r} = \langle 6, 5, 1 \rangle$, which is a point on L . We can find one

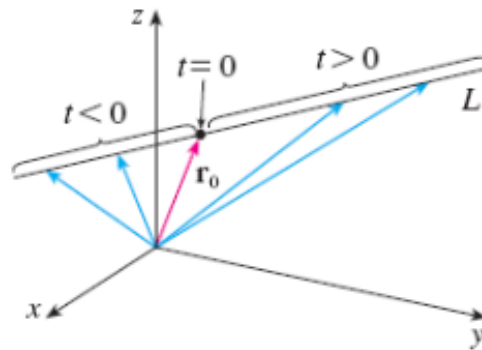


Figure 3.0.3: Figure for Example 3.0.3

more point on L by taking $t = -1$ in a similar way.

□

4 Equation of a straight line in space passing through 2 points

In the previous section, we had seen that the information given to us (for finding the equation of a straight line L) was the following:

- A fixed point P_0 on the straight line L .
- A vector $\vec{v}$ to which the straight line L is parallel to.

But now, we will be given 2 points $P_0(x_0, y_0, z_0)$ and $P_1(x_1, y_1, z_1)$ on the straight line L instead.

With the information about one more extra point P_1 on the straight line, we will find out a vector $\vec{v}$ parallel to the straight line. If we consider the vector $\overrightarrow{P_0P_1}$, then this vector $\overrightarrow{P_0P_1}$ lies on the straight line (whose equation we are seeking). This means that the vector $\overrightarrow{P_0P_1}$ itself is parallel to the straight line. So we can take $\vec{v}$ to be equal to the vector $\overrightarrow{P_0P_1}$.

Take a variable point $P(x, y, z)$ on the straight line. See Figure 4.0.1 for an illustration.

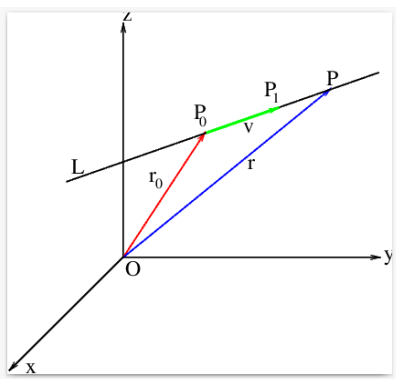


Figure 4.0.1: Straight line passing through two points

Remark 4.0.1. It doesn't matter where we take the variable point P on the straight line. We can take this point P to be anywhere on the straight line L . It doesn't have to be on the right hand side of the points P_0 and P_1 .

Join P_0, P_1 and P with the origin O . Let $\vec{r} = \overrightarrow{OP}$, $\vec{r}_1 = \overrightarrow{OP_1}$ and $\vec{r}_0 = \overrightarrow{OP_0}$. It is then easy to see that

$$\overrightarrow{OP} = \overrightarrow{OP_0} + \overrightarrow{P_0P}.$$

But clearly the vector $\overrightarrow{P_0P}$ is a scalar multiple of the vector $\overrightarrow{P_0P_1}$. So we have:

$$\vec{r} = \vec{r}_0 + t\overrightarrow{P_0P_1},$$

for some scalar t . Since $\overrightarrow{P_0P_1} = \vec{r}_1 - \vec{r}_0$, we have (from the previous equation):

$$\vec{r} = \vec{r}_0 + t(\vec{r}_1 - \vec{r}_0).$$

Or, in other words,

$$\vec{r} = (1 - t)\vec{r}_0 + t\vec{r}_1,$$

where t is a real scalar.

Question: What is the equation of the portion of the straight line L that lies between P_0 and P_1 ?

Answer: Basically, this means that we have to find out the locus of any point on the line segment P_0P_1 . The equation of the entire straight line is given by:

$$\vec{r} = (1 - t)\vec{r}_0 + t\vec{r}_1,$$

where t is a real scalar. Then for the portion P_0P_1 of the same straight line, the value of the parameter t should be restricted. It is easy to see that the value of the parameter t should then lie between 0 and 1 (The value $t = 0$ gives the vector $\vec{r}_0$, which corresponds to the point P_0 and the value $t = 1$ corresponds to the point P_1 in a similar way.). So the equation of the portion P_0P_1 of the same line is given by:

$$\vec{r} = (1 - t)\vec{r}_0 + t\vec{r}_1, \quad 0 \leq t \leq 1.$$

Remark 4.0.2. In many books, only the equation of the portion P_0P_1 of the straight line is given. If it is only this portion, then $0 \leq t \leq 1$. If it is the entire straight line, then t is any real number.

5 Equation of a plane

When we were trying to find out the equation of a straight line in space, we had the information about a fixed point P_0 on the straight line and a vector

$\vec{v}$ (which is known) to which the straight line is parallel to. Basically, the vector $\vec{v}$ determined the direction of the straight line.

Now, when we come to finding the vector equation of a plane, definitely we are going to take a fixed point lying on the plane (as an information given to us). But what about the direction of the plane? We can have infinite number of straight lines in the plane passing through the particular fixed point! All those lines define infinitely many directions. But we need to find a unique direction, which is going to give us some idea about the direction and the equation of the plane. This unique direction will be given by a **normal** to the plane. If you know about a normal to the plane at one point, then by parallel displacement, you can make that normal vector to be a normal at any point on that plane.

So the information that will be given to us (for finding the equation of a plane) is the following:

- a fixed point $P_0 (x_0, y_0, z_0)$ lying on the plane.
- the normal $\vec{n}$ to the plane at that particular point.

Please see Figure 5.0.1 for an illustration.

Let $P (x, y, z)$ be a variable point on the plane. Let O denote the origin.

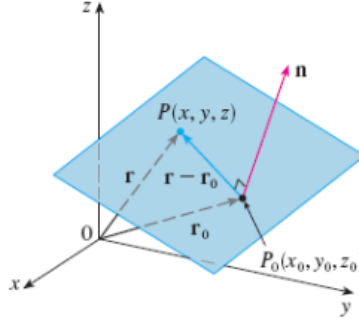


Figure 5.0.1: A plane in space

Join OP and OP_0 . Let $\overrightarrow{OP} = \vec{r}$ and $\overrightarrow{OP_0} = \vec{r_0}$. Then $\overrightarrow{P_0P} = \vec{r} - \vec{r_0}$ is a vector lying on the plane. Clearly then, the vector $\vec{r} - \vec{r_0}$ is perpendicular to the vector $\vec{n}$. So, from the property of dot product, we get:

$$(\vec{r} - \vec{r_0}) \cdot \vec{n} = 0.$$

If $\vec{n} = \langle a, b, c \rangle$, then the above equation becomes:

$$\langle x - x_0, y - y_0, z - z_0 \rangle \cdot \langle a, b, c \rangle = 0.$$

This implies that

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0.$$

This is the required **equation of the plane**. The above equation can be rewritten as:

$$ax + by + cz = d, \quad (*)$$

where $d = ax_0 + by_0 + cz_0$.

Remark 5.0.1. Whenever you see an equation of a plane in the form $(*)$, you can say that the components of the normal to the plane are the coefficients of x, y and z in the equation $(*)$.

Remark 5.0.2. If we need to find the **angle between two planes**, then we have to draw the normals to those two planes. The angle between those two normals is going to be equal to the angle between the two planes. See Figure 5.0.2 for an illustration.

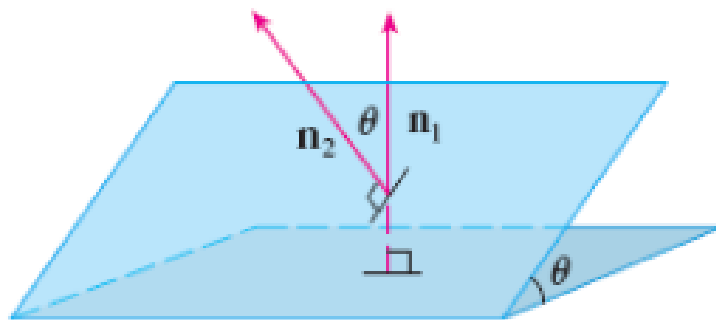


Figure 5.0.2: Angle between two planes

Example 5.0.3. Find the equation of the plane that passes through the points $P(1, 3, 2)$, $Q(3, -1, 6)$ and $R(5, 2, 0)$. See Figure 5.0.3 for an illustration. We can choose any one of P, Q, R as the fixed point on the plane. Let us choose it to be P . [It doesn't matter which point (out of P, Q, R) we

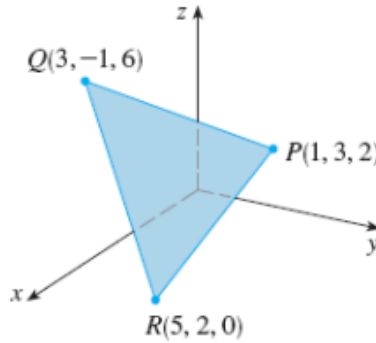


Figure 5.0.3: Figure for Example 5.0.3

choose as our fixed point on the plane. Whatever we choose, we are going to end up with the same equation of the plane!]

Now we need to find out the components of the normal to the plane at the point P . For doing this, consider the two vectors $\overrightarrow{PQ}$ and $\overrightarrow{PR}$. Then $\overrightarrow{PQ} = \langle 2, -4, 4 \rangle$ and $\overrightarrow{PR} = \langle 4, -1, -2 \rangle$. If we consider the *cross product* of $\overrightarrow{PQ}$ and $\overrightarrow{PR}$, we will get a vector which is perpendicular to both of them. But both the vectors $\overrightarrow{PQ}$ and $\overrightarrow{PR}$ lie on the same plane. So the cross product vector is going to be normal to the plane at the point P . Let $\vec{n}$ denote the cross product vector. Then we have:

$$\begin{aligned}\vec{n} &= \langle 2, -4, 4 \rangle \times \langle 4, -1, -2 \rangle \\ &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -4 & 4 \\ 4 & -1 & -2 \end{vmatrix} = 12\hat{i} + 20\hat{j} + 14\hat{k}.\end{aligned}$$

Now we have $P = (1, 3, 2)$ as the fixed point on the plane and $\vec{n} = \langle 12, 20, 14 \rangle$ as the normal to the plane at P . So the equation of the plane will be:

$$12(x - 1) + 20(y - 3) + 14(z - 2) = 0.$$

That is,

$$6x + 10y + 7z = 50.$$

If you take Q or R as the fixed point, and an appropriate combination of two vectors on the plane, then you will get the same answer after simplification (You can check this!). $\square$

Example 5.0.4. Find the angle between the two planes $x + y + z = 1$ and $x - 2y + 3z = 1$. Also, find the symmetric equation of the line L of intersection of these two planes. See Figure 5.0.4 for an illustration.

Let π_1 and π_2 denote the planes $x + y + z = 1$ and $x - 2y + 3z = 1$ respectively.

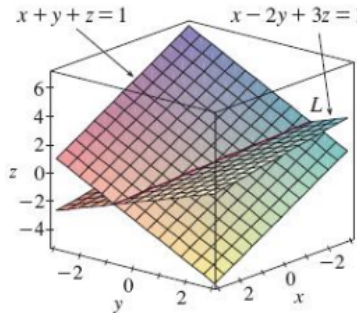


Figure 5.0.4: Figure for Example 5.0.4

Let $\vec{n}_1$ and $\vec{n}_2$ denote the normals to the planes π_1 and π_2 respectively. It then follows from Remark 5.0.1 that $\vec{n}_1 = \langle 1, 1, 1 \rangle$ and $\vec{n}_2 = \langle 1, -2, 3 \rangle$. If θ is the angle between these two planes, then θ is also the angle between the vectors $\vec{n}_1$ and $\vec{n}_2$. Hence we have:

$$\vec{n}_1 \cdot \vec{n}_2 = |\vec{n}_1| |\vec{n}_2| \cos \theta.$$

That is, $1 - 2 + 3 = \sqrt{3}\sqrt{14}\cos \theta$. So, $\theta = \cos^{-1}(\frac{2}{\sqrt{42}})$.

The given two planes π_1 and π_2 are not parallel (because the components of their normals are not scalar multiples of each other). Two planes which are not parallel are going to intersect in a line. We now need to find the symmetric equation of the line L of intersection of the planes π_1 and π_2 . To find the equation of L , we need a fixed point on it, and we need the direction ratios of L . The line L lies on both the planes π_1 and π_2 . So L must be perpendicular to both $\vec{n}_1$ and $\vec{n}_2$.

On the other hand, if $\vec{n} = \vec{n}_1 \times \vec{n}_2$, then $\vec{n}$ is going to be perpendicular to both $\vec{n}_1$ and $\vec{n}_2$. That means, the directions of L and $\vec{n}$ are the same. So L and $\vec{n}$ are going to have proportional direction ratios. Note that

$$\vec{n} = \vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 1 & -2 & 3 \end{vmatrix} = 5\hat{i} - 2\hat{j} - 3\hat{k}.$$

So $\vec{n}$ is a vector parallel to the line L .

Now, we have to find out a point on the line L . For that, first observe that the line L intersects the xy -plane. That is, L is not parallel to the xy -plane. This is because the direction ratios of L are proportional to $\langle 5, -2, -3 \rangle$ (which are the direction ratios of $\vec{n}$). And any line parallel to the xy -plane will have direction ratios proportional to $\langle 1, 1, 0 \rangle$ (**because:** any line parallel to the xy -plane will make angles $0, 0, \pi/2$ with the x, y and z -axes respectively, and $\cos(0) = 1, \cos(\frac{\pi}{2}) = 0$). But $\langle 1, 1, 0 \rangle$ and $\langle 5, -2, -3 \rangle$ are not scalar multiples of each other!

Then the point of intersection of L with the xy -plane must have its z -component 0. Putting $z = 0$ in the equations of the two planes, we get:

$$x + y = 1$$

$$x - 2y = 1.$$

Solving these equations, we get, $x = 1, y = 0$. So, $(1, 0, 0)$ is a point on L . It is in fact, the point of intersection of L with the xy -plane. Hence the symmetric equation of the line L is given by:

$$\frac{x-1}{5} = \frac{y-0}{-2} = \frac{z-0}{-3}.$$

Figure 5.0.5 illustrates this. □

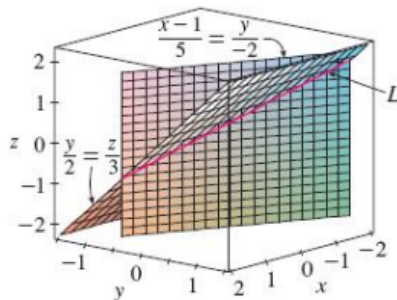


Figure 5.0.5: Figure for Example 5.0.4

6 Vector functions

Recall the definition of a **function**: Consider two sets A and B . If to each element of A , we can uniquely assign (via a rule) an element of B , and if we call that rule f , then we say that f is a **function** from the set A to the set B .

Now consider an interval $I = [a, b]$ in the real line $\mathbb{R}$. Let f, g, h be some functions from I to $\mathbb{R}$. Then corresponding to every point $t \in I$, we are going to get the values $f(t), g(t), h(t)$ in $\mathbb{R}$. Let V_3 denote the set of all 3 dimensional vectors. Let $\vec{r}$ be a particular vector in V_3 . Clearly then, $\vec{r}$ has 3 components. Corresponding to each $t \in I$, if we choose these 3 components as $f(t), g(t), h(t)$, then in the 3 dimensional space, we will be able to define a vector $\vec{r}(t)$, which is given by:

$$\vec{r}(t) = \langle f(t), g(t), h(t) \rangle. \quad (1)$$

So in this way, we have defined a function from I into V_3 . The values of this function $\vec{r}(t)$ depend upon the value of $t \in I$. Such a function is called a **vector function** or a **vector-valued function**.

In terms of the unit vectors $\hat{i}, \hat{j}, \hat{k}$ along the x, y, z -axes respectively, we can write $\vec{r}(t)$ as:

$$\vec{r}(t) = f(t)\hat{i} + g(t)\hat{j} + h(t)\hat{k}.$$

Example 6.0.1. $\vec{r}(t) = \langle \sin t, 2 - \cos t, t \rangle$ (for all $t \in [0, \pi]$) is a vector function. Observe that $\sin t$, $2 - \cos t$ and t are all functions from $[0, \pi]$ to $\mathbb{R}$. If we define $f(t) = \sin t$, $g(t) = 2 - \cos t$, and $h(t) = t$, then $\vec{r}(t)$ is of the form given by equation (1). $\square$

6.1 Limit of a vector function

Suppose $\vec{r}(t)$ is a vector function, where t belongs to some interval $I = [\alpha, \beta]$. Let $a \in I$ and $\vec{b} \in V_3$ be a fixed vector.

Definition 6.1.1. We say that the **limit** of $\vec{r}(t)$ as t tends to a equals $\vec{b}$ if given any $\epsilon > 0$, there exists a $\delta > 0$ (ϵ and δ are very small positive real numbers) such that

$$|\vec{r}(t) - \vec{b}| < \epsilon \text{ whenever } 0 < |t - a| < \delta.$$

We denote this by $\lim_{t \rightarrow a} \vec{r}(t) = \vec{b}$. $\square$

Remark 6.1.2. Please see the analogy of the above definition with the concept of limit of a function of a single variable (which you have learnt in the first half of this course MA 101).

Remark 6.1.3. If $\vec{r}(t) = \langle f(t), g(t), h(t) \rangle$ and $\vec{b} = \langle b_1, b_2, b_3 \rangle$, then since

$$\lim_{t \rightarrow a} \vec{r}(t) = \langle \lim_{t \rightarrow a} f(t), \lim_{t \rightarrow a} g(t), \lim_{t \rightarrow a} h(t) \rangle,$$

we get that:

$$\lim_{t \rightarrow a} f(t) = b_1$$

$$\lim_{t \rightarrow a} g(t) = b_2$$

$$\lim_{t \rightarrow a} h(t) = b_3.$$

Exercise: Show that $\lim_{t \rightarrow a} \vec{r}(t) = \vec{b}$ if and only if for every $\epsilon > 0$, there exists a number $\delta > 0$ such that if $0 < |t - a| < \delta$, then $|\vec{r}(t) - \vec{b}| < \epsilon$.

A remark about the exercise: The “if” part of the above exercise is the definition of limit itself. We only need to show that “only if” part of the exercise. Note that this exercise gives a necessary and sufficient condition for the existence of a limit.

Proof of the exercise: We will now prove the “only if” part.

Let us assume that $\lim_{t \rightarrow a} \vec{r}(t) = \vec{b}$. Let us also assume that $\vec{r}(t) = \langle f(t), g(t), h(t) \rangle$ and $\vec{b} = \langle b_1, b_2, b_3 \rangle$. Then we have:

$$\lim_{t \rightarrow a} f(t) = b_1$$

$$\lim_{t \rightarrow a} g(t) = b_2 \quad (2)$$

$$\lim_{t \rightarrow a} h(t) = b_3.$$

This set (2) of equations defines a limit for each of the functions f, g and h at the point a .

If we go by the definition of limit for a function of one variable, the equations in the set (2) imply that: Given any $\epsilon > 0$, there exists positive real numbers $\delta_1, \delta_2, \delta_3$ such that

$$|f(t) - b_1| < \epsilon \text{ whenever } 0 < |t - a| < \delta_1$$

$$|g(t) - b_2| < \epsilon \text{ whenever } 0 < |t - a| < \delta_2 \quad (*)$$

$$|h(t) - b_3| < \epsilon \text{ whenever } 0 < |t - a| < \delta_3.$$

Now we are going to do some mathematical manipulations. In (*) above, observe that it doesn't matter if we replace ϵ by $\frac{\epsilon}{2}$ or $\frac{\epsilon}{3}$ or $\frac{\epsilon}{100}$ or something similar. This is because we want the expressions $|f(t) - b_1|$, $|g(t) - b_2|$, $|h(t) - b_3|$ to be as close to zero as possible! So in (*), we can replace ϵ by $\frac{\epsilon}{3}$. Recall what we are supposed to prove: We are supposed to prove that given the limit $\lim_{t \rightarrow a} \vec{r}(t)$ exists and equals $\vec{b}$, we must have $|\vec{r}(t) - \vec{b}| < \epsilon$ whenever $0 < |t - a| < \delta$ for some $\delta > 0$. Now, if we take $\delta = \min(\delta_1, \delta_2, \delta_3)$, and replace $\delta_1, \delta_2, \delta_3$ in (*) by δ , then we get that the following holds true:

$$|f(t) - b_1| < \frac{\epsilon}{3} \text{ whenever } 0 < |t - a| < \delta$$

$$|g(t) - b_2| < \frac{\epsilon}{3} \text{ whenever } 0 < |t - a| < \delta \quad (**)$$

$$|h(t) - b_3| < \frac{\epsilon}{3} \text{ whenever } 0 < |t - a| < \delta.$$

Now if we look at the expression $|\vec{r}(t) - \vec{b}|$ and look at the components of the vector $\vec{r}(t) - \vec{b}$, we will get that:

$$|\vec{r}(t) - \vec{b}| = \sqrt{|f(t) - b_1|^2 + |g(t) - b_2|^2 + |h(t) - b_3|^2}.$$

It is a general fact that for any 3 positive real numbers a, b, c , we have:

$$\sqrt{|a|^2 + |b|^2 + |c|^2} \leq |a| + |b| + |c|.$$

Hence we get:

$$\begin{aligned} |\vec{r}(t) - \vec{b}| &= \sqrt{|f(t) - b_1|^2 + |g(t) - b_2|^2 + |h(t) - b_3|^2} \\ &\leq |f(t) - b_1| + |g(t) - b_2| + |h(t) - b_3| \\ &< \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} \text{ whenever } 0 < |t - a| < \delta \\ &= \epsilon \text{ whenever } 0 < |t - a| < \delta. \end{aligned}$$

So we proved what we wanted. □

6.2 Continuity of a vector function at a point

Definition 6.2.1. Let $I = [\alpha, \beta]$ be an interval in $\mathbb{R}$. Let a be a fixed point in I . We say that a vector function $\vec{r}(t)$ is **continuous at a** if

$$\lim_{t \rightarrow a} \vec{r}(t) = \vec{r}(a).$$

□

7 Space curves

Recall from the previous section that when we were considering the functions f, g and h , we were considering them as a function of a single variable t . We could have used any symbol other than t . But we intentionally use the symbol t because it generally represents “time”.

Let $\vec{r}(t) = \langle f(t), g(t), h(t) \rangle$ be a vector function. For any fixed value of t , we can relate $\vec{r}(t)$ to the point in the 3 dimensional space whose coordinates are $(f(t), g(t), h(t))$.

The interval I on which the vector function $\vec{r}(t)$ is defined, is generally (in the real world) taken as $[0, t_F]$, where t_F is some real number > 0 . This is because if t denotes time, then we are going to consider something that is happening from time $t = 0$ to a subsequent time.

Now, suppose we are trying to track the motion of a particle in space. If at each time t , we trace the position of the particle, then we are going to get a curve in the space. That means, if for all values of t in the interval $[\alpha, \beta]$, we are going to consider the location of the particle. The set of all points in space corresponding to the location of the particle at different times t ($t \in [\alpha, \beta]$), is going to generate a curve C . And for each point on the curve C , one can define a vector (see Figure 7.0.1). So a **space curve** is constructed in this way.

Remark 7.0.1. Observe that by varying t , we are extracting a curve C in the above mentioned procedure. Therefore, t is a parameter. That's why, a space curve is also known as a **parametric curve**.

Example 7.0.2. Suppose we have a curve given by

$$\vec{r}(t) = \langle \cos t, \sin t, t \rangle, \quad t \in [0, 2\pi].$$

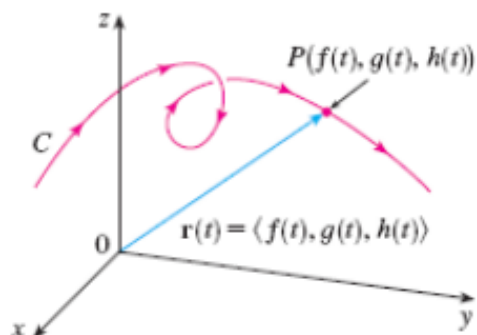


Figure 7.0.1: A space curve

Now if we consider the 3 dimensional space, then at time $t = 0$, the location of the particle in the 3 dimensional space is given by $(\cos 0, \sin 0, 0) = (1, 0, 0)$. Similarly, at $t = \frac{\pi}{2}$, we get the point $(0, 1, \frac{\pi}{2})$ in space. If we consider the first 2 components of $\vec{r}(t)$, they are given by $\cos t$ and $\sin t$. Since $\cos^2 t + \sin^2 t = 1$, therefore it will define a circle of radius 1. For the vector function $\vec{r}(t)$ in this example, we are going to obtain a curve as shown in Figure 7.0.2. This is the way we can construct space curves. $\square$

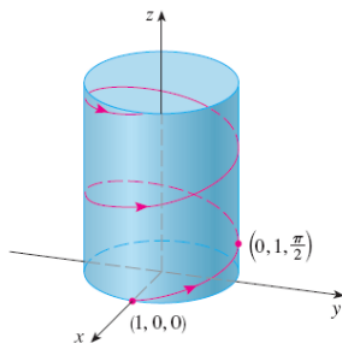


Figure 7.0.2: The space curve for Example 7.0.2

Example 7.0.3. Describe the curve defined by the vector function

$$\vec{r}(t) = \langle 1 + t, 2 + 5t, -1 + 6t \rangle .$$

If we look at the components of $\vec{r}(t)$, we can see that they are in a specific form (namely, the form of an equation of a straight line). The curve is therefore going to be a straight line passing through the point $(1, 2, -1)$ and the straight line will have $(1, 5, 6)$ as direction ratios. See Figure 7.0.3. $\square$

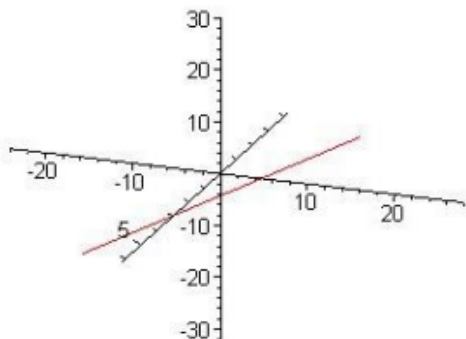


Figure 7.0.3: The space curve for Example 7.0.3

Now suppose a 3 dimensional curve is given in terms of x, y, z , not in terms of the parameter t . There are some ways to parametrize such a curve. For this, please see the example below.

Example 7.0.4. Find a vector function that represents the curve of intersection of the cylinder $x^2 + y^2 = 1$ and the plane $y + z = 2$. Observe that if the cylinder $x^2 + y^2 = 1$ is cut by a plane parallel to the xy -plane, then the curve of intersection is a circle of radius 1. But here the plane is $y + z = 2$, which is not parallel to the xy -plane. So the curve of intersection is not going to be a circle of radius 1. It is going to be an ellipse in this case, see Figure 7.0.4 for an illustration. But how can we parametrize this curve (ellipse)?

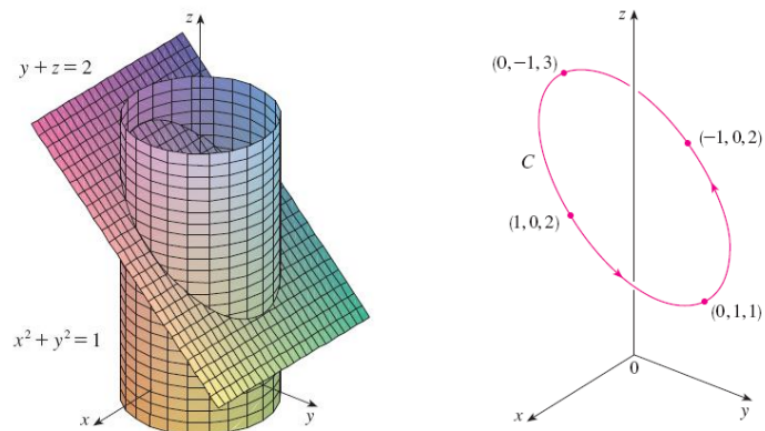


Figure 7.0.4: The curve of intersection for Example 7.0.4

Look at the equation of the cylinder. If we use the parameter t , then $x^2 + y^2 = 1$ can be easily seen to be parametrized by $x = \cos t, y = \sin t$. So we can write:

$$\vec{r}(t) = \langle \cos t, \sin t, 2 - \sin t \rangle, \text{ where } t \in [0, 2\pi].$$

□

—————END—————